

MA 301

Test 2

Fall - 2011

Problem	points	out of
1		10
2		10
3		10
4		6
5		12
Score		48

Name: _____

HR: _____

Signature: _____

Formulas

* Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$

- $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

- $\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)] \quad (z \neq 0)$

* Let $z = r(\cos \theta + i \sin \theta)$

- $z^n = r^n [\cos(n\theta) + i \sin(n\theta)]$

- $z^{1/n} = r^{1/n} \left[\cos \left(\frac{\theta + 2\pi k}{n} \right) + i \sin \left(\frac{\theta + 2\pi k}{n} \right) \right]$

for $k = 0, 1, 2, \dots, n - 1$

Problem 1: Let $f(x) = \begin{cases} 0 & -2 < t < -1 \\ 4 & -1 < t < 1 \\ 0 & 1 < t < 2 \end{cases}$ and $f(x) = f(x + 4)$

(a) Graph $f(x)$ on the interval $[-3, 5]$

(b) Find the Fourier Series of $f(x)$

Problem 2: Let $f(x) = -x$ on $[-1, 1]$ and satisfies $f(x) = f(x + 2)$

(a) Graph $f(x)$ on the interval $[-3, 3]$

(b) Find the Fourier Series of $f(x)$

Problem 3: Let $f(x) = \begin{cases} 1 & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$ and $f(x) = f(x + 2\pi)$

(a) Graph $f(x)$ on the interval $[-2\pi, 2\pi]$

(b) Find the Fourier Series of $f(x)$

Problem 4: Let $z_1 = 5 + i$, $z_2 = 2 - 4i$, $z_3 = a + bi$.

Perform the following operations and write the result in standard form.

(a) $z_1 + z_2$

(b) $z_1 z_3$

(c) $\frac{z_2}{z_1}$

(d) $|z_2|$

(e) $z_1 - iz_2$

(f) $\sqrt{-10} \cdot \sqrt{-6}$

Problem 5: Let $z_1 = 1 - i$, $z_2 = 3i$. Answer the following:

(a) Find the polar form of z_1 and plot it.

(b) Find the polar form of z_2 and plot it.

(c) Find z_1^{12} .

(d) Find $z_1^{1/3}$.

(d) Solve for z : $z^4 + z_1 = z_2$